

COMPASS_Synoptic_SEAL_Data_Analysis_Sept2022

Stephanie J. Wilson

2023-03-03

##Run notes

```
Run_notes = "GWI_202209_TR_LysC_10cm was duplicated without a dilution for Vanadium NOx. Appears to be :  
measured concentrations are essentially equal, removed second run (duplicate) in this code."
```

Information

```
#####  
##### COMPASS Synoptic  
##### Chesapeake Bay Sites  
##### Data Analysis Code: Porewater Nutrients  
##### MONTH: August 2022  
#####  
  
##### Information #####  
#Author: Stephanie J. Wilson  
#Edited: 20220727  
  
#Samples taken from Lysimeters & Sippers at CB Synoptic Sites  
#Samples were filtered with 0.45 uM, kept on ice, frozen until analysis  
#Field Protocol:  
#Samples Analyzed on a SEAL discrete auto analyzer  
#Lab Protocol: https://docs.google.com/document/d/1VaJT7Wb9AcdmM1tgsR\_9ZtQ6kwcaoNmp/edit?usp=sharing&ou  
#NOx method = https://drive.google.com/file/d/1sicqBFnzVxmDd5I2\_pu8s8pj7iNOAuhF/view?usp=sharing  
#NH4 method = https://drive.google.com/file/d/1ENGemUEvm\_rffZqv3lz9BjD0pAMX5nzu/view?usp=sharing  
#PO4 method = https://drive.google.com/file/d/1m3gXDZnJoIo\_QmyhvuZG4HRgGShCzm9Wq/view?usp=sharing  
#Units from SEAL = mg/L and converted to uMoles/L
```

QAQC on Slopes

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
library(data.table)
```

```
#read in datafile with all the slopes
```

```
qlog <- read.csv("Raw Data/SERC_SEAL_STDs_Log.csv")
```

```
head(qlog)
```

```
##      Date      User  Machine Analysis  Slope Intercept      R2
## 1 20220513 Stephanie Wilson SERC SEAL    NH3 1.9370    -0.030 0.9995
## 2 20220513 Stephanie Wilson SERC SEAL    P04 2.8690      0.002 0.9994
## 3 20220525 Stephanie Wilson SERC SEAL  V-Nox 1.0000      0.000 0.9990
## 4 20220614 Stephanie Wilson SERC SEAL    NH3 1.1487    -0.011 0.9999
## 5 20220614 Stephanie Wilson SERC SEAL    P04 2.3950    -0.001 0.9998
## 6 20220615 Stephanie Wilson SERC SEAL  V-Nox 1.0000      0.000 0.9998
##   Nox_Red_Eff X
## 1             NA
## 2             NA
## 3             NA
## 4             NA
## 5             NA
## 6             NA
```

```
#pull out each method
```

```
qNH3 <- qlog[qlog$Analysis %like% "NH3", ]
```

```
head(qNH3)
```

```
##      Date      User  Machine Analysis  Slope Intercept      R2
## 1 20220513 Stephanie Wilson SERC SEAL    NH3 1.9370    -0.030 0.9995
## 4 20220614 Stephanie Wilson SERC SEAL    NH3 1.1487    -0.011 0.9999
## 7 20220615 Stephanie Wilson SERC SEAL    NH3 1.7070    -0.008 1.0000
## 10 20220718 Stephanie Wilson SERC SEAL    NH3 1.7080    -0.017 0.9995
## 12 20220718 Stephanie Wilson SERC SEAL    NH3 1.6590    -0.017 0.9997
## 14 20220719 Stephanie Wilson SERC SEAL    NH3 1.5560      0.000 0.9994
##   Nox_Red_Eff X
## 1             NA
## 4             NA
## 7             NA
## 10            NA
## 12            NA
## 14            NA
```

```
qP04 <- qlog[qlog$Analysis %like% "P04", ]
```

```
head(qP04)
```

```
##      Date      User  Machine Analysis  Slope Intercept      R2
## 2 20220513 Stephanie Wilson SERC SEAL    P04 2.869      0.002 0.9994
## 5 20220614 Stephanie Wilson SERC SEAL    P04 2.395    -0.001 0.9998
## 8 20220615 Stephanie Wilson SERC SEAL    P04 2.349      0.000 0.9999
## 11 20220718 Stephanie Wilson SERC SEAL    P04 2.445    -0.003 0.9993
## 13 20220718 Stephanie Wilson SERC SEAL    P04 2.349    -0.004 0.9992
## 15 20220719 Stephanie Wilson SERC SEAL    P04 2.271    -0.003 0.9993
##   Nox_Red_Eff X
## 2             NA
```

```
## 5      NA
## 8      NA
## 11     NA
## 13     NA
## 15     NA
```

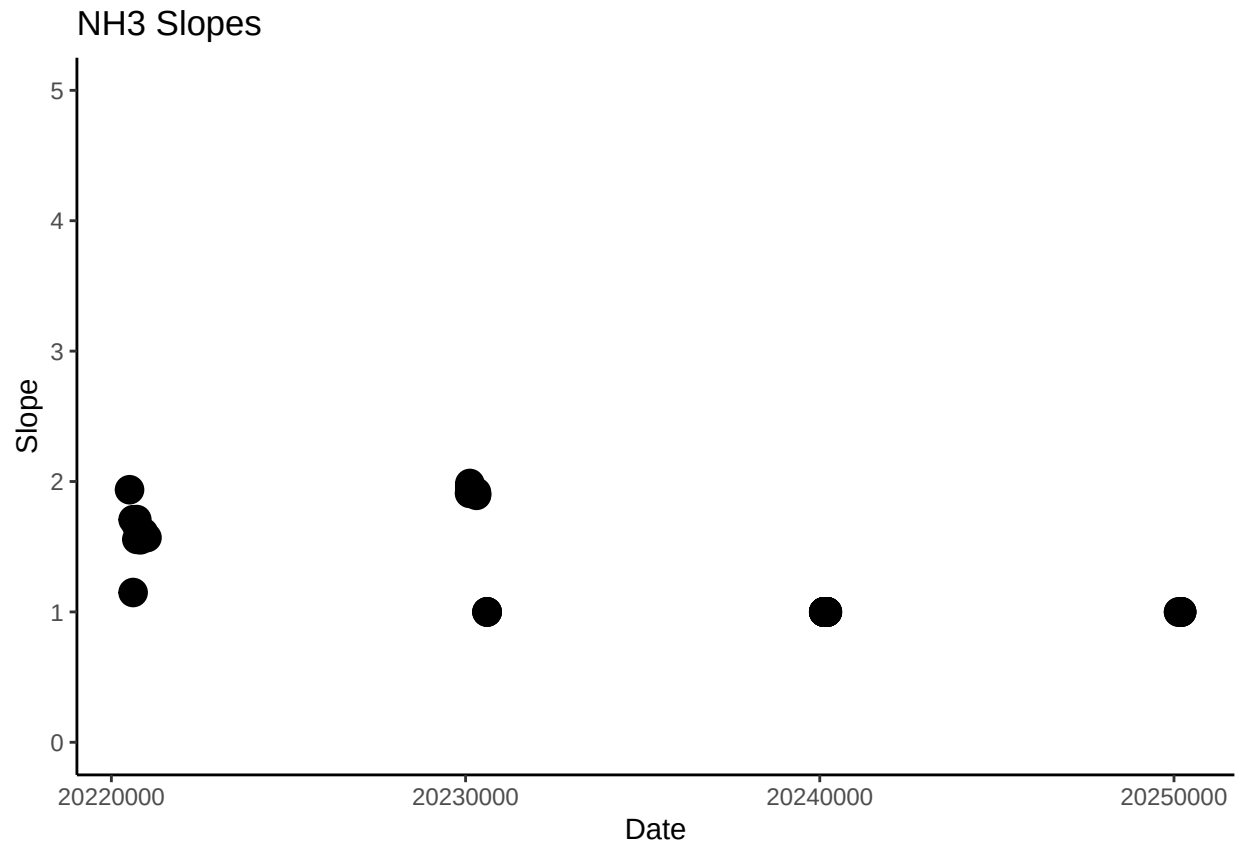
```
qNOx <- qlog[qlog$Analysis %like% "V-Nox", ]
head(qNOx)
```

```
##      Date      User  Machine Analysis Slope Intercept      R2
## 3  20220525 Stephanie Wilson  SERC  SEAL    V-Nox      1      0 0.9990
## 6  20220615 Stephanie Wilson  SERC  SEAL    V-Nox      1      0 0.9998
## 9  20220718 Stephanie Wilson  SERC  SEAL    V-Nox      1      0 0.9999
## 16 20220730 Stephanie Wilson  SERC  SEAL    V-Nox      1      0 1.0000
## 17 20220731 Stephanie Wilson  SERC  SEAL    V-Nox      1      0 1.0000
## 22 20220824 Stephanie Wilson  SERC  SEAL    V-Nox      1      0 0.9988
##      Nox_Red_Eff X
## 3      NA
## 6      NA
## 9      NA
## 16     NA
## 17     NA
## 22     NA
```

```
##### NH3
#plot the slopes to make sure there are no crazy outliers
slope1 <- ggplot(data=qNH3, aes(x=Date, y=Slope)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0, 5) +
  theme(legend.position="none") +
  ggtitle("NH3 Slopes")

slope1
```

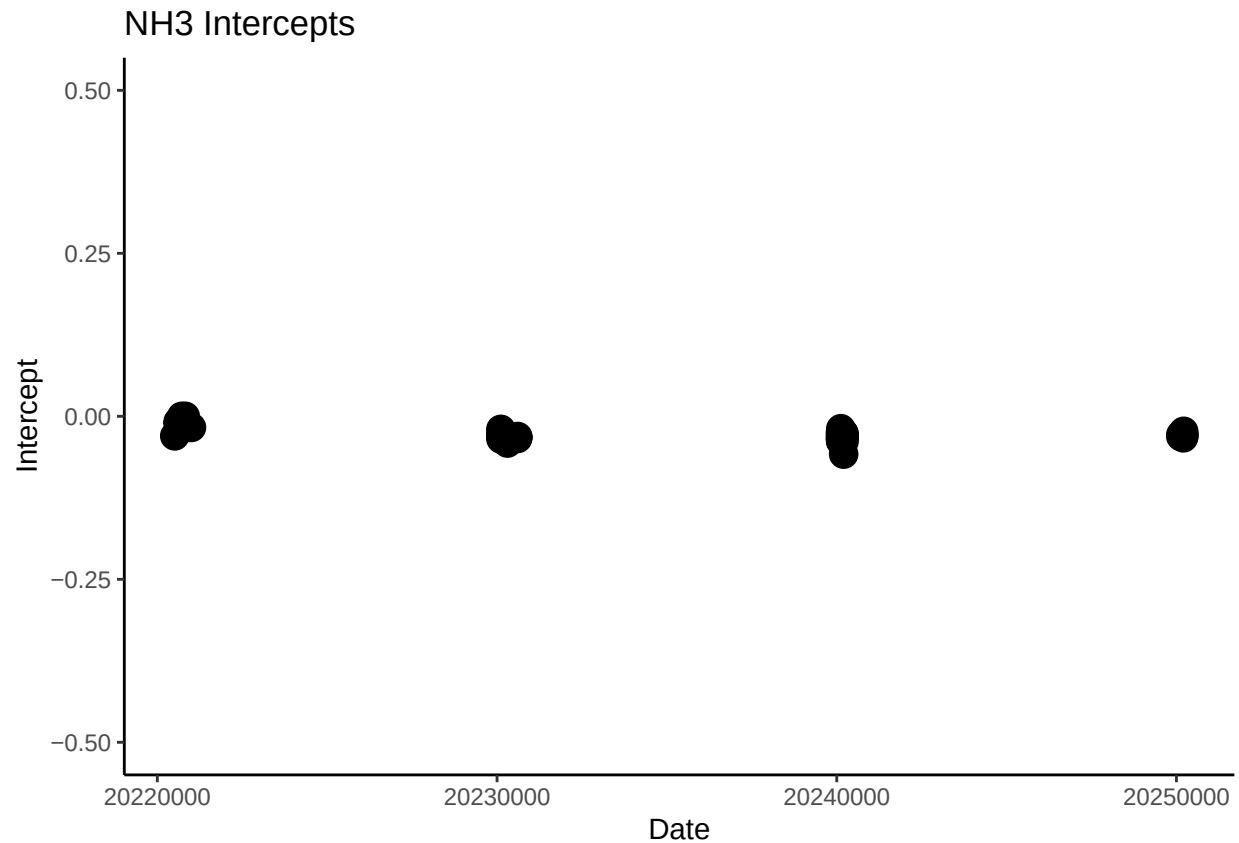
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the intercepts to make sure there are no crazy outliers
int1 <- ggplot(data=qNH3, aes(x=Date, y=Intercept)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(-0.5,0.5) +
  theme(legend.position="none")+
  ggtitle("NH3 Intercepts")

int1
```

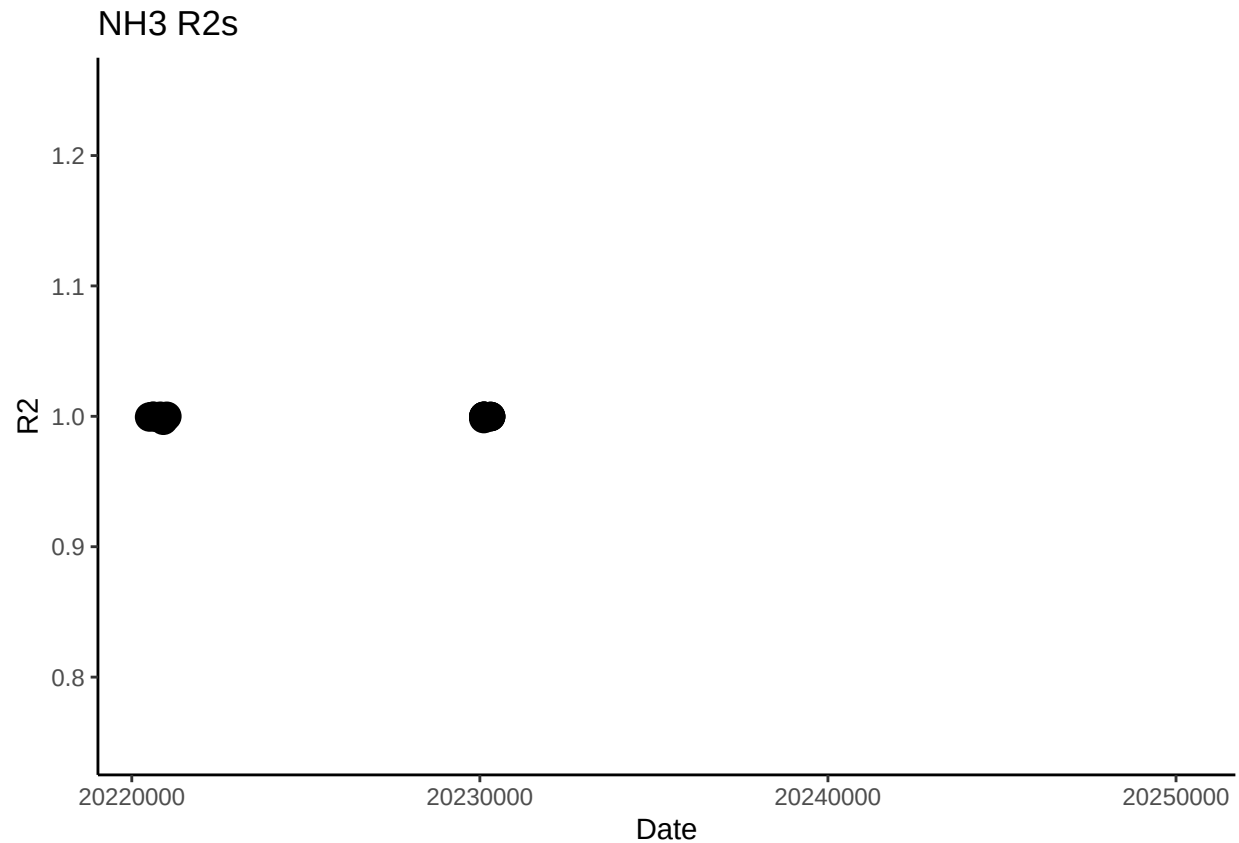
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the R2s to make sure there are no crazy outliers
Rsqr1 <- ggplot(data=qNH3, aes(x=Date, y=R2)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0.75, 1.25) +
  theme(legend.position="none")+
  ggtitle("NH3 R2s")
```

Rsqr1

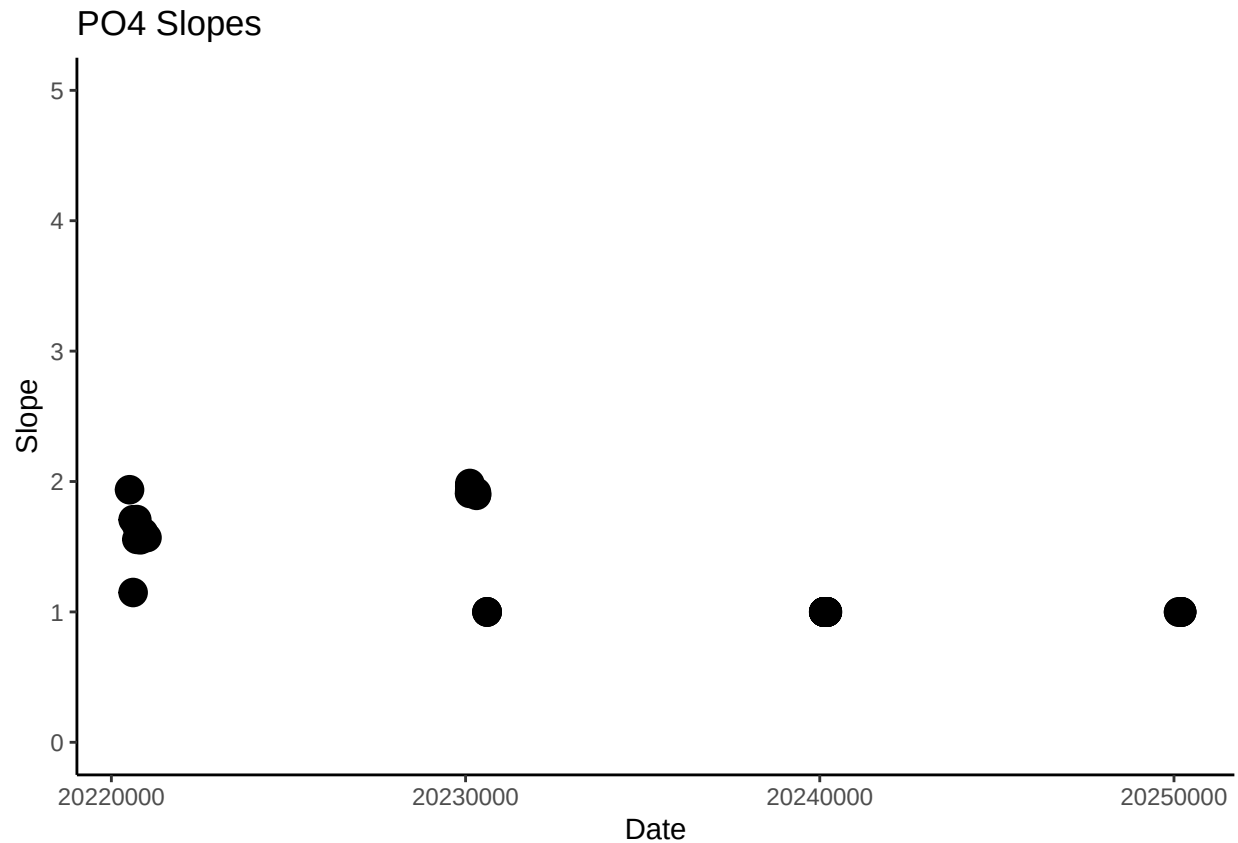
```
## Warning: Removed 39 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
##### P04
slope2 <- ggplot(data=qNH3, aes(x=Date, y=Slope)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0, 5) +
  theme(legend.position="none")+
  ggtitle("P04 Slopes")

slope2
```

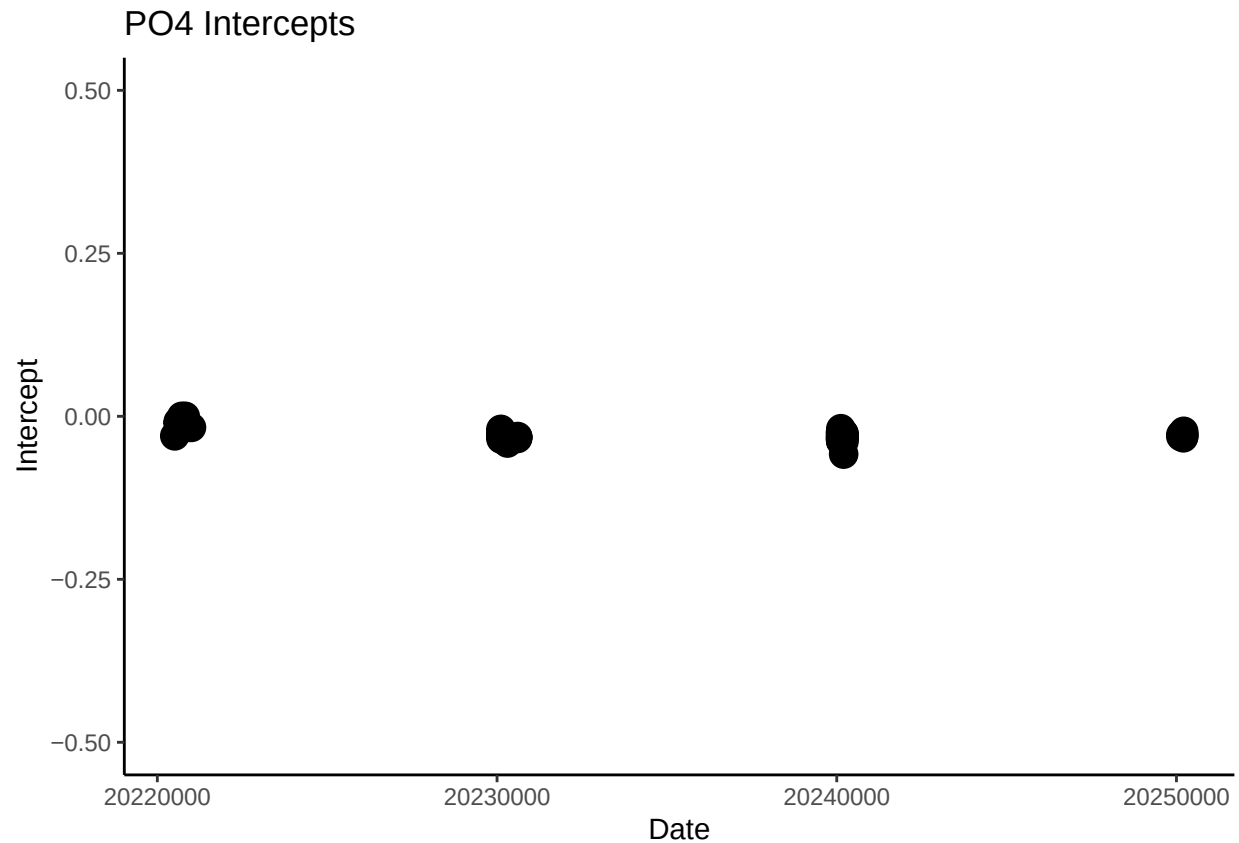
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the intercepts to make sure there are no crazy outliers
int2 <- ggplot(data=qNH3, aes(x=Date, y=Intercept)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(-0.5,0.5) +
  theme(legend.position="none")+
  ggtitle("P04 Intercepts")

int2
```

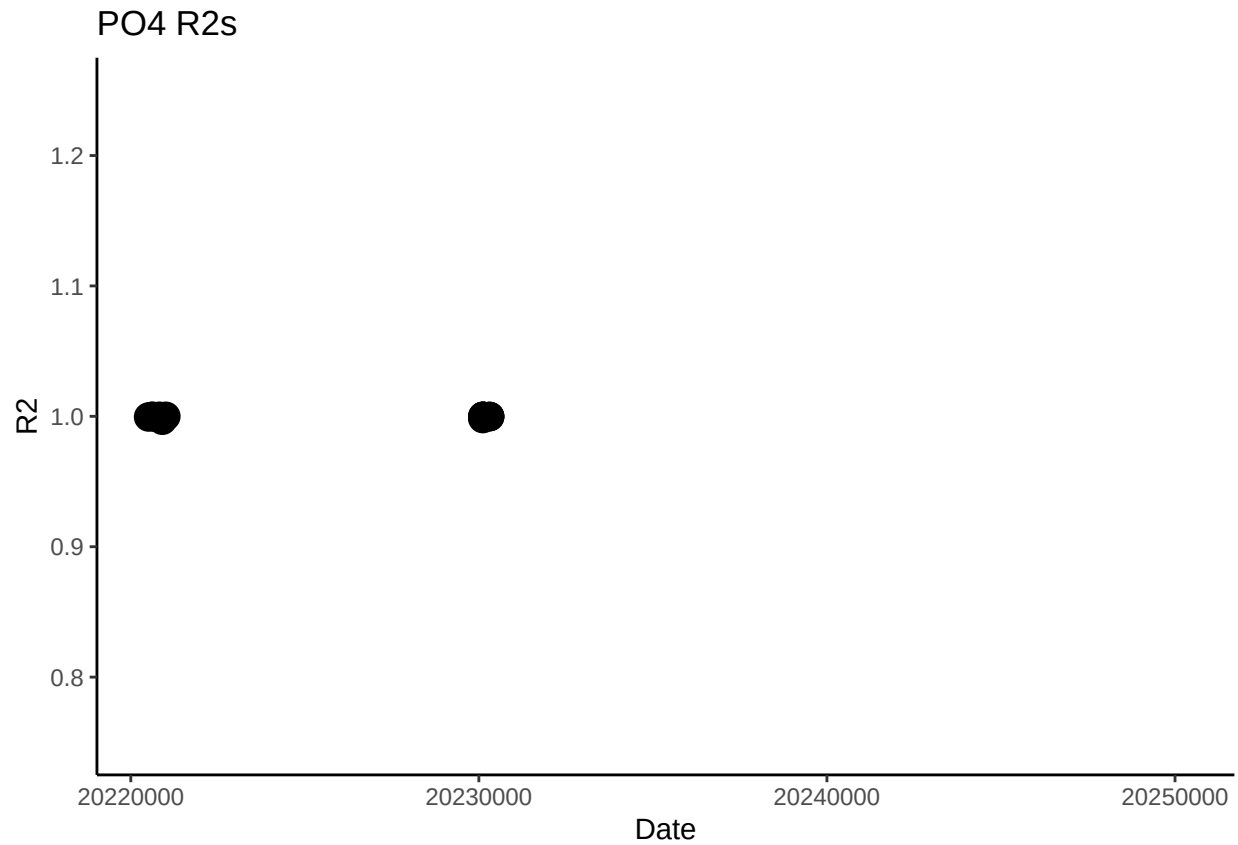
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the R2s to make sure there are no crazy outliers
Rsqr2 <- ggplot(data=qNH3, aes(x=Date, y=R2)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0.75, 1.25) +
  theme(legend.position="none")+
  ggtitle("PO4 R2s")
```

Rsqr2

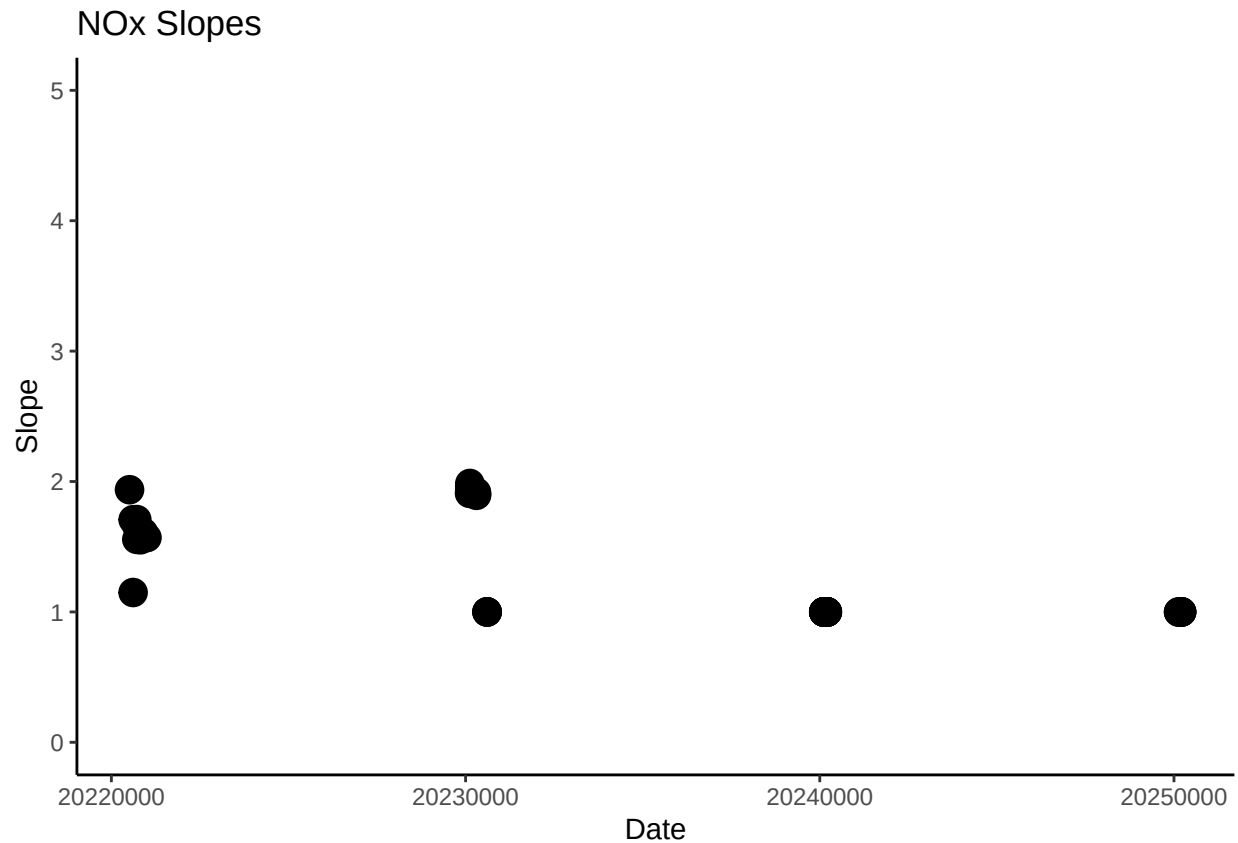
```
## Warning: Removed 39 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
##### NOx
slope3 <- ggplot(data=qNH3, aes(x=Date, y=Slope)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0, 5) +
  theme(legend.position="none")+
  ggtitle("NOx Slopes")

slope3
```

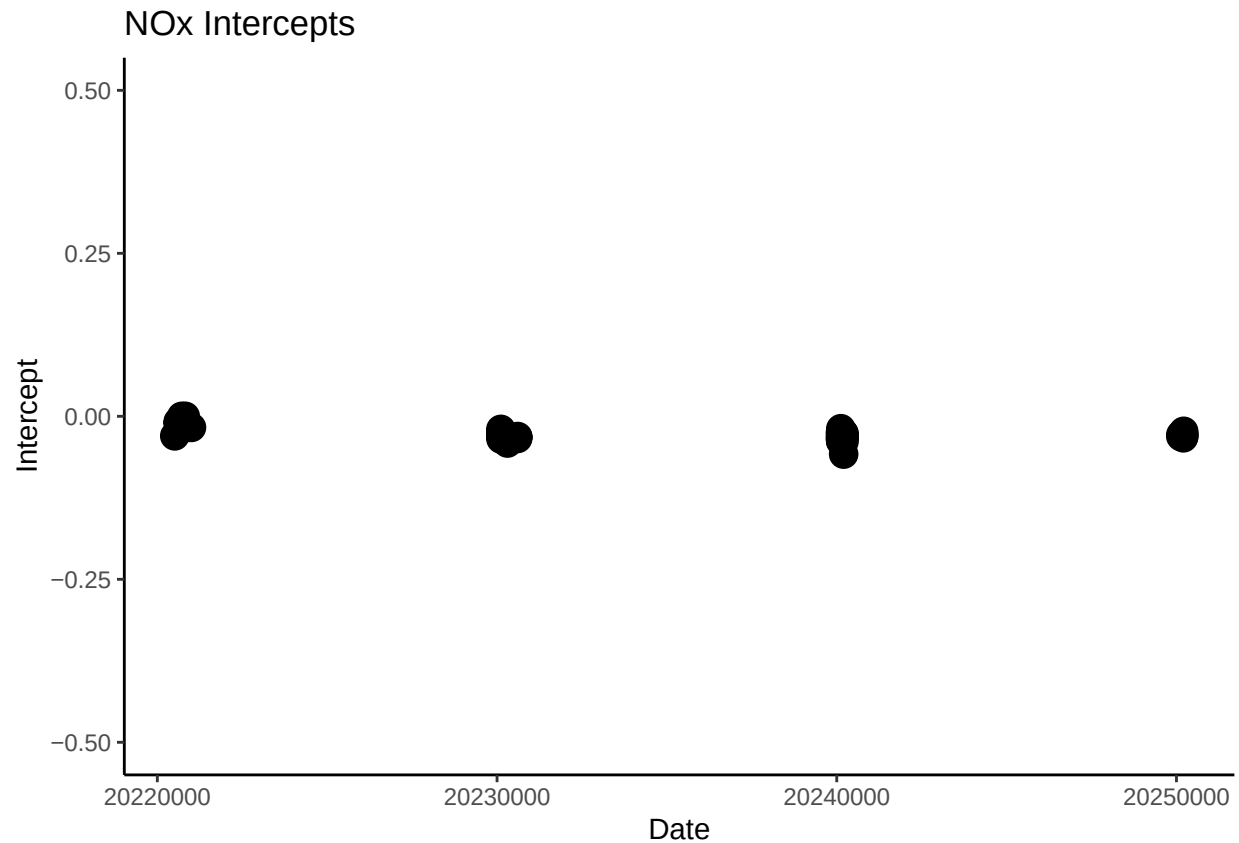
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the intercepts to make sure there are no crazy outliers
int3 <- ggplot(data=qNH3, aes(x=Date, y=Intercept)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(-0.5,0.5) +
  theme(legend.position="none")+
  ggtitle("NOx Intercepts")

int3
```

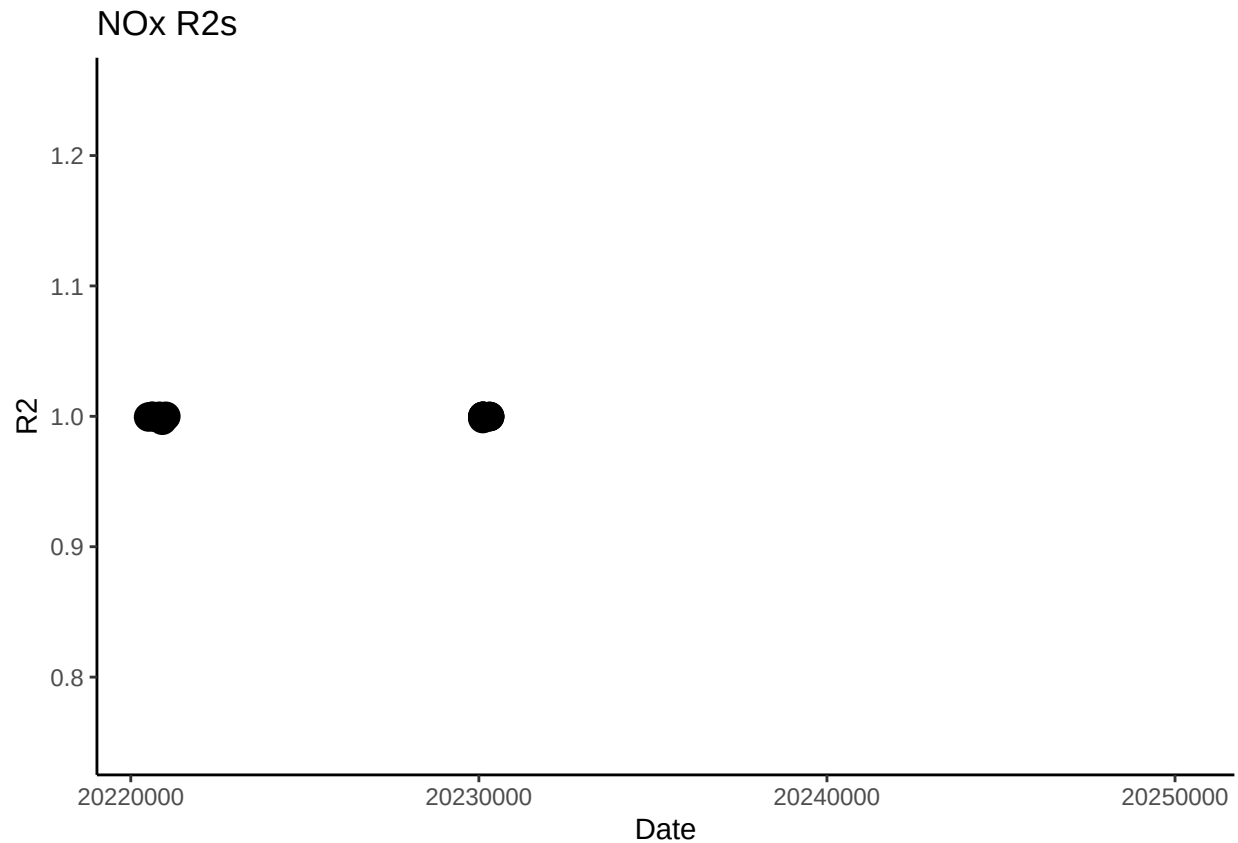
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
#plot the R2s to make sure there are no crazy outliers
Rsqr3 <- ggplot(data=qNH3, aes(x=Date, y=R2)) +
  #geom_line()+
  geom_point(aes(size=3)) +
  theme_classic() + ylim(0.75, 1.25) +
  theme(legend.position="none")+
  ggtitle("NOx R2s")
```

Rsqr3

```
## Warning: Removed 39 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Code Set up

```
#packages:
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:data.table':
##
##   between, first, last

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(data.table)
library(matrixStats)
```

```
##
## Attaching package: 'matrixStats'

## The following object is masked from 'package:dplyr':
##
## count
```

```
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:dplyr':
##
## combine
```

```
library(ggpubr)
library(grid)
```

Ammonia & Phosphate

```
#read in data
file1 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202209_1.csv")
#Quick look at dataframe
head(file1)
```

```
## RUNSTARTED X1553789462 X3.28.2019.12.11 X X.1 X.2 X.3 X.4
## 1 RESULT -1 S1 Standard 1 0 0.000019 0.000019 0
## 2 RESULT -2 S90 Standard 90 1 0.008778 0.008778 0
## 3 RESULT -2 S91 Standard 91 2 0.023144 0.023144 0
## 4 RESULT -2 S92 Standard 92 3 0.045439 0.045439 0
## 5 RESULT -2 S93 Standard 93 4 0.117195 0.117195 0
## 6 RESULT -2 S94 Standard 94 5 0.237585 0.237585 0
## X.5 X.6 X.7 X.8 X.9 X.10 X.11
## 1 0 0 0 mg N/L Nitrite 0.2 1553790057 3/28/2019 12:20
## 2 0 0 0 mg N/L Nitrite 0.2 1553790098 3/28/2019 12:21
## 3 0 0 0 mg N/L Nitrite 0.2 1553790139 3/28/2019 12:22
## 4 0 0 0 mg N/L Nitrite 0.2 1553790179 3/28/2019 12:22
## 5 0 0 0 mg N/L Nitrite 0.2 1553790220 3/28/2019 12:23
## 6 0 0 0 mg N/L Nitrite 0.2 1553790261 3/28/2019 12:24
```

```
#take out only the columns that we need
dat1 <- file1[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(dat1) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
dat1$Run_Number <- "1"
head(dat1)
```

##	Run_Info	Sample_Name	Conc	Abs	Dilution	Units	Test	Run_Number
## 1	RESULT	Standard 1	0.000019	0.000019		0 mg N/L Nitrite 0.2		1
## 2	RESULT	Standard 90	0.008778	0.008778		0 mg N/L Nitrite 0.2		1
## 3	RESULT	Standard 91	0.023144	0.023144		0 mg N/L Nitrite 0.2		1
## 4	RESULT	Standard 92	0.045439	0.045439		0 mg N/L Nitrite 0.2		1
## 5	RESULT	Standard 93	0.117195	0.117195		0 mg N/L Nitrite 0.2		1
## 6	RESULT	Standard 94	0.237585	0.237585		0 mg N/L Nitrite 0.2		1

```
#read in data
file2 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202209_2.csv")
#Quick look at dataframe
head(file2)
```

##	RUNSTARTED	X1568815852	X9.18.2019.10.10	X	X.1	X.2	X.3	X.4
## 1	RESULT	-1	S1	Standard	1	0 0.584576	0.584576	0
## 2	RESULT	-2	S90	Standard	90	1 0.518989	0.518989	0
## 3	RESULT	-2	S91	Standard	91	2 0.605939	0.605939	0
## 4	RESULT	-2	S92	Standard	92	3 0.684434	0.684434	0
## 5	RESULT	-2	S93	Standard	93	4 0.787007	0.787007	0
## 6	RESULT	-2	S94	Standard	94	5 0.938437	0.938437	0

##	X.5	X.6	X.7	X.8	X.9	X.10	X.11
## 1	0	0	0 mg N/L	NO3_NO2	2	1568817080	9/18/2019 10:31
## 2	0	0	0 mg N/L	NO3_NO2	2	1568817209	9/18/2019 10:33
## 3	0	0	0 mg N/L	NO3_NO2	2	1568817338	9/18/2019 10:35
## 4	0	0	0 mg N/L	NO3_NO2	2	1568817466	9/18/2019 10:37
## 5	0	0	0 mg N/L	NO3_NO2	2	1568817594	9/18/2019 10:39
## 6	0	0	0 mg N/L	NO3_NO2	2	1568817722	9/18/2019 10:42

```
#take out only the columns that we need
dat2 <- file2[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(dat2) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
dat2$Run_Number <- "2"
head(dat2)
```

##	Run_Info	Sample_Name	Conc	Abs	Dilution	Units	Test	Run_Number
## 1	RESULT	Standard 1	0.584576	0.584576		0 mg N/L NO3_NO2 2		2
## 2	RESULT	Standard 90	0.518989	0.518989		0 mg N/L NO3_NO2 2		2
## 3	RESULT	Standard 91	0.605939	0.605939		0 mg N/L NO3_NO2 2		2
## 4	RESULT	Standard 92	0.684434	0.684434		0 mg N/L NO3_NO2 2		2
## 5	RESULT	Standard 93	0.787007	0.787007		0 mg N/L NO3_NO2 2		2
## 6	RESULT	Standard 94	0.938437	0.938437		0 mg N/L NO3_NO2 2		2

```
#read in data
file3 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NH3_PO4_202209_3.csv")
#Quick look at dataframe
head(file3)
```

##	RUNSTARTED	X1673644940	X1.13.2023.16.22	X	X.1	X.2	X.3
## 1	RESULT	-1	S1	Standard	1	0 0.009933	0.009933
## 2	RESULT	-2	S90	Standard	.0389	1 0.029260	0.029260
## 3	RESULT	-2	S91	Standard	.1000	2 0.063726	0.063726

```
## 4      RESULT      -2          S92 Standard .2000   3 0.094039 0.094039
## 5      RESULT      -2          S93 Standard .5000   4 0.271131 0.271131
## 6      RESULT      -2          S94 Standard 1.0000   5 0.558844 0.558844
##   X.4 X.5 X.6 X.7   X.8       X.9       X.10       X.11
## 1    0    0    0    0 mg N/L Ammonia 2 1673646166 1/13/2023 16:42
## 2    0    0    0    0 mg N/L Ammonia 2 1673646326 1/13/2023 16:45
## 3    0    0    0    0 mg N/L Ammonia 2 1673646486 1/13/2023 16:48
## 4    0    0    0    0 mg N/L Ammonia 2 1673646646 1/13/2023 16:50
## 5    0    0    0    0 mg N/L Ammonia 2 1673646806 1/13/2023 16:53
## 6    0    0    0    0 mg N/L Ammonia 2 1673646966 1/13/2023 16:56
```

```
#take out only the columns that we need
dat3 <- file3[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(dat3) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
dat3$Run_Number <- "3"
head(dat3)
```

```
##   Run_Info      Sample_Name      Conc      Abs Dilution Units      Test
## 1   RESULT      Standard 1 0.009933 0.009933          0 mg N/L Ammonia 2
## 2   RESULT Standard .0389 0.029260 0.029260          0 mg N/L Ammonia 2
## 3   RESULT Standard .1000 0.063726 0.063726          0 mg N/L Ammonia 2
## 4   RESULT Standard .2000 0.094039 0.094039          0 mg N/L Ammonia 2
## 5   RESULT Standard .5000 0.271131 0.271131          0 mg N/L Ammonia 2
## 6   RESULT Standard 1.0000 0.558844 0.558844          0 mg N/L Ammonia 2
##   Run_Number
## 1           3
## 2           3
## 3           3
## 4           3
## 5           3
## 6           3
```

```
alldat <- rbind(dat1, dat2, dat3)

#Pull out standards
stds <- alldat[alldat$Sample_Name %like% "Standard", ]
head(stds)
```

```
##   Run_Info Sample_Name      Conc      Abs Dilution Units      Test Run_Number
## 1   RESULT Standard 1 0.000019 0.000019          0 mg N/L Nitrite 0.2          1
## 2   RESULT Standard 90 0.008778 0.008778          0 mg N/L Nitrite 0.2          1
## 3   RESULT Standard 91 0.023144 0.023144          0 mg N/L Nitrite 0.2          1
## 4   RESULT Standard 92 0.045439 0.045439          0 mg N/L Nitrite 0.2          1
## 5   RESULT Standard 93 0.117195 0.117195          0 mg N/L Nitrite 0.2          1
## 6   RESULT Standard 94 0.237585 0.237585          0 mg N/L Nitrite 0.2          1
```

```
#Pull out samples
alldat2 <- alldat[alldat$Sample_Name %like% "MSM_", ]
alldat2 <- rbind(alldat2, (alldat[alldat$Sample_Name %like% "GWI_", ]))
alldat2 <- rbind(alldat2, (alldat[alldat$Sample_Name %like% "GCrew_", ]))
head(alldat2)
```

```
##      Run_Info      Sample_Name      Conc      Abs Dilution Units
## 122  RESULT MSM_202209_UP_LysA_20cm 1.390346 0.896168      0 mg N/L
## 123  RESULT MSM_202209_UP_LysA_45cm 0.681030 0.444370      0 mg N/L
## 124  RESULT MSM_202209_UP_LysB_10cm 0.029888 0.029626      0 mg N/L
## 125  RESULT MSM_202209_UP_LysB_20cm 0.194465 0.134453      0 mg N/L
## 126  RESULT MSM_202209_UP_LysB_45cm 0.395437 0.262462      0 mg N/L
## 127  RESULT MSM_202209_UP_LysC_10cm 0.375479 0.249749      0 mg N/L
##      Test Run_Number
## 122 Ammonia 2      1
## 123 Ammonia 2      1
## 124 Ammonia 2      1
## 125 Ammonia 2      1
## 126 Ammonia 2      1
## 127 Ammonia 2      1
```

```
alldat3 <- alldat2
```

NO_x

```
#read in data
Nfile1 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NOx_202209_1.csv")
#Quick look at dataframe
head(Nfile1)
```

```
##      RUNSTARTED X1674261103      X1.20.2023.19.31      X X.1      X.2      X.3
## 1      RUNENDED 1674261139 INSUFFICIENT REAGENT      NA      NA      NA
## 2 RUNSTARTED 1674261195      1/20/2023 19:33      NA      NA      NA
## 3      RESULT      -1      S1 Standard 1      0 0.184456 0.184456
## 4      RESULT      -2      S90 Standard 90      1 0.203401 0.203401
## 5      RESULT      -2      S91 Standard 91      2 0.227204 0.227204
## 6      RESULT      -2      S92 Standard 92      3 0.266972 0.266972
##      X.4 X.5 X.6 X.7      X.8      X.9      X.10      X.11
## 1      NA NA NA NA      NA      NA      NA
## 2      NA NA NA NA      NA      NA      NA
## 3      0 0 0 512 mg N/L Vanadium NOx 1674263383 1/20/2023 20:09
## 4      0 0 0 512 mg N/L Vanadium NOx 1674263465 1/20/2023 20:11
## 5      0 0 0 512 mg N/L Vanadium NOx 1674263546 1/20/2023 20:12
## 6      0 0 0 512 mg N/L Vanadium NOx 1674263627 1/20/2023 20:13
```

```
#take out only the columns that we need
Ndat1 <- Nfile1[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(Ndat1) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
Ndat1$Run_Number <- "1"
head(Ndat1)
```

```
##      Run_Info Sample_Name      Conc      Abs Dilution Units      Test
## 1      RUNENDED      NA      NA      NA
## 2 RUNSTARTED      NA      NA      NA
## 3      RESULT Standard 1 0.184456 0.184456      0 mg N/L Vanadium NOx
## 4      RESULT Standard 90 0.203401 0.203401      0 mg N/L Vanadium NOx
```

```
## 5      RESULT Standard 91 0.227204 0.227204      0 mg N/L Vanadium NOx
## 6      RESULT Standard 92 0.266972 0.266972      0 mg N/L Vanadium NOx
## Run_Number
## 1      1
## 2      1
## 3      1
## 4      1
## 5      1
## 6      1
```

```
#read in data
Nfile2 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NOx_202209_2.csv")
#Quick look at dataframe
head(Nfile2)
```

```
## RUNSTARTED X1673807323 X1.15.2023.13.28      X X.1      X.2      X.3 X.4
## 1      RESULT      -1      S1 Standard 1      0 0.140496 0.140496      0
## 2      RESULT      -2      S90 Standard 90      1 0.158302 0.158302      0
## 3      RESULT      -2      S91 Standard 91      2 0.182799 0.182799      0
## 4      RESULT      -2      S92 Standard 92      3 0.223878 0.223878      0
## 5      RESULT      -2      S93 Standard 93      4 0.344134 0.344134      0
## 6      RESULT      -2      S94 Standard 94      5 0.553945 0.553945      0
## X.5 X.6 X.7      X.8      X.9      X.10      X.11
## 1      0      0      0 mg N/L Vanadium NOx 1673809505 1/15/2023 14:05
## 2      0      0      0 mg N/L Vanadium NOx 1673809585 1/15/2023 14:06
## 3      0      0      0 mg N/L Vanadium NOx 1673809665 1/15/2023 14:07
## 4      0      0      0 mg N/L Vanadium NOx 1673809745 1/15/2023 14:09
## 5      0      0      0 mg N/L Vanadium NOx 1673809825 1/15/2023 14:10
## 6      0      0      0 mg N/L Vanadium NOx 1673809905 1/15/2023 14:11
```

```
#There is a duplicate in this run for no reason, we remove the 2nd one here because the numbers are ess
Nfile2 <- Nfile2 %>%
  filter(!X1.15.2023.13.28 == "U28")

#take out only the columns that we need
Ndat2 <- Nfile2[,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(Ndat2) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
Ndat2$Run_Number <- "2"
head(Ndat2)
```

```
## Run_Info Sample_Name      Conc      Abs Dilution Units      Test
## 1      RESULT Standard 1 0.140496 0.140496      0 mg N/L Vanadium NOx
## 2      RESULT Standard 90 0.158302 0.158302      0 mg N/L Vanadium NOx
## 3      RESULT Standard 91 0.182799 0.182799      0 mg N/L Vanadium NOx
## 4      RESULT Standard 92 0.223878 0.223878      0 mg N/L Vanadium NOx
## 5      RESULT Standard 93 0.344134 0.344134      0 mg N/L Vanadium NOx
## 6      RESULT Standard 94 0.553945 0.553945      0 mg N/L Vanadium NOx
## Run_Number
## 1      2
## 2      2
## 3      2
## 4      2
```

```
## 5      2
## 6      2
```

```
#read in data
Nfile3 <- read.csv("Raw Data/SEAL_COMPASS_Synoptic_NOx_202209_3.csv")
#Quick look at dataframe
head(Nfile3)
```

```
##  RUNSTARTED X1673913818 X1.16.2023.19.03      X X.1      X.2      X.3 X.4
## 1 RUNSTARTED 1673918924 1/16/2023 20:28      NA      NA      NA  NA
## 2  RESULT      -1      S1 Standard 1  0 0.139612 0.139612  0
## 3  RESULT      -2      S90 Standard 90 1 0.156868 0.156868  0
## 4  RESULT      -2      S91 Standard 91 2 0.177989 0.177989  0
## 5  RESULT      -2      S92 Standard 92 3 0.220473 0.220473  0
## 6  RESULT      -2      S93 Standard 93 4 0.339950 0.339950  0
##  X.5 X.6 X.7      X.8      X.9      X.10      X.11
## 1  NA  NA  NA      NA      NA
## 2  0  0  0 mg N/L Vanadium NOx 1673921121 1/16/2023 21:05
## 3  0  0  0 mg N/L Vanadium NOx 1673921200 1/16/2023 21:06
## 4  0  0  0 mg N/L Vanadium NOx 1673921281 1/16/2023 21:08
## 5  0  0  0 mg N/L Vanadium NOx 1673921360 1/16/2023 21:09
## 6  0  0  0 mg N/L Vanadium NOx 1673921441 1/16/2023 21:10
```

```
#take out only the columns that we need
Ndat3 <- Nfile3[ ,c(1,4,6,7, 9, 12, 13)]
# assigning new names to the columns of the data frame
colnames(Ndat3) <- c('Run_Info','Sample_Name','Conc', "Abs", "Dilution", "Units", "Test")
Ndat3$Run_Number <- "3"
head(Ndat3)
```

```
##  Run_Info Sample_Name      Conc      Abs Dilution Units      Test
## 1 RUNSTARTED      NA      NA      NA
## 2  RESULT Standard 1 0.139612 0.139612      0 mg N/L Vanadium NOx
## 3  RESULT Standard 90 0.156868 0.156868      0 mg N/L Vanadium NOx
## 4  RESULT Standard 91 0.177989 0.177989      0 mg N/L Vanadium NOx
## 5  RESULT Standard 92 0.220473 0.220473      0 mg N/L Vanadium NOx
## 6  RESULT Standard 93 0.339950 0.339950      0 mg N/L Vanadium NOx
##  Run_Number
## 1      3
## 2      3
## 3      3
## 4      3
## 5      3
## 6      3
```

```
Nalldat <- rbind(Ndat1, Ndat2, Ndat3)

#Pull out standards
Nstds <- Nalldat[Nalldat$Sample_Name %like% "Standard", ]
head(Nstds)
```

```
##  Run_Info Sample_Name      Conc      Abs Dilution Units      Test
```

```
## 3   RESULT   Standard 1 0.184456 0.184456      0 mg N/L Vanadium NOx
## 4   RESULT   Standard 90 0.203401 0.203401      0 mg N/L Vanadium NOx
## 5   RESULT   Standard 91 0.227204 0.227204      0 mg N/L Vanadium NOx
## 6   RESULT   Standard 92 0.266972 0.266972      0 mg N/L Vanadium NOx
## 7   RESULT   Standard 93 0.382567 0.382567      0 mg N/L Vanadium NOx
## 8   RESULT   Standard 94 0.583302 0.583302      0 mg N/L Vanadium NOx
```

```
##   Run_Number
## 3           1
## 4           1
## 5           1
## 6           1
## 7           1
## 8           1
```

```
#Pull out samples
```

```
Nalldat2 <- Nalldat[Nalldat$Sample_Name %like% "MSM_", ]
Nalldat2 <- rbind(Nalldat2, (Nalldat[Nalldat$Sample_Name %like% "GWI_", ]))
Nalldat2 <- rbind(Nalldat2, (Nalldat[Nalldat$Sample_Name %like% "GCrew_", ]))
head(Nalldat2)
```

```
##   Run_Info      Sample_Name      Conc      Abs Dilution  Units
## 23  RESULT MSM_202209_UP_LysA_20cm 0.005178 0.189194      0 mg N/L
## 24  RESULT MSM_202209_UP_LysA_45cm 0.004450 0.188606      0 mg N/L
## 25  RESULT MSM_202209_UP_LysB_10cm -0.000600 0.184527      0 mg N/L
## 26  RESULT MSM_202209_UP_LysB_20cm -0.000584 0.184540      0 mg N/L
## 27  RESULT MSM_202209_UP_LysB_45cm 0.020174 0.201302      0 mg N/L
## 28  RESULT MSM_202209_UP_LysC_10cm -0.002939 0.182637      0 mg N/L
##           Test Run_Number
## 23 Vanadium NOx      1
## 24 Vanadium NOx      1
## 25 Vanadium NOx      1
## 26 Vanadium NOx      1
## 27 Vanadium NOx      1
## 28 Vanadium NOx      1
```

```
Nalldat3 <- Nalldat2
```

Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections
```

```
## Duplicated samples: MSM_202209_WC_LysC_45cm, MSM_202209_UP_LysC_10cm, MSM_202209_TR_LysC_45cm, MSM_202209_WC_LysC_10cm
```

```
##
```

```
## All duplicated samples have valid dilutions. No naming issues detected.
```

Dilution Corrections NOx - ensure the latest dilution is kept

```
## Dilution Corrections
```

```
## No duplicated samples.
```

Constants

```
N_mw <- 14.0067 # molecular weight of N
P_mw <- 30.973762 # molecular weight of P
Con1 <- 1000 # conversion factor value
Con2 <- 1000000 # conversion factor value
```

Convert Data from mg/L to uM

```
head(alldat2)
```

```
## # A tibble: 6 x 10
##   Run_Info Sample_Name      Conc      Abs Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>      <dbl> <dbl>    <dbl> <chr> <chr> <chr>    <chr>
## 1 RESULT    MSM_202209_UP_~ 1.39  0.896      1 mg N~ Ammo~ 1      MSM_20~
## 2 RESULT    MSM_202209_UP_~ 0.681 0.444      1 mg N~ Ammo~ 1      MSM_20~
## 3 RESULT    MSM_202209_UP_~ 0.0299 0.0296      1 mg N~ Ammo~ 1      MSM_20~
## 4 RESULT    MSM_202209_UP_~ 0.194 0.134      1 mg N~ Ammo~ 1      MSM_20~
## 5 RESULT    MSM_202209_UP_~ 0.395 0.262      1 mg N~ Ammo~ 1      MSM_20~
## 6 RESULT    MSM_202209_UP_~ 0.375 0.250      1 mg N~ Ammo~ 1      MSM_20~
## # i 1 more variable: Original_Name <chr>
```

```
head(Nalldat2)
```

```
## # A tibble: 6 x 10
##   Run_Info Sample_Name      Conc      Abs Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>      <dbl> <dbl>    <dbl> <chr> <chr> <chr>    <chr>
## 1 RESULT    MSM_202209_UP~ 5.18e-3 0.189      1 mg N~ Vana~ 1      MSM_20~
## 2 RESULT    MSM_202209_UP~ 4.45e-3 0.189      1 mg N~ Vana~ 1      MSM_20~
## 3 RESULT    MSM_202209_UP~ -6 e-4 0.185      1 mg N~ Vana~ 1      MSM_20~
## 4 RESULT    MSM_202209_UP~ -5.84e-4 0.185      1 mg N~ Vana~ 1      MSM_20~
## 5 RESULT    MSM_202209_UP~ 2.02e-2 0.201      1 mg N~ Vana~ 1      MSM_20~
## 6 RESULT    MSM_202209_UP~ -2.94e-3 0.183      1 mg N~ Vana~ 1      MSM_20~
## # i 1 more variable: Original_Name <chr>
```

```
#subset by test
```

```
NH4samples <- subset(alldat2, Test == "Ammonia 2")
```

```
head(NH4samples)
```

```
## # A tibble: 6 x 10
##   Run_Info Sample_Name      Conc      Abs Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>      <dbl> <dbl>    <dbl> <chr> <chr> <chr>    <chr>
## 1 RESULT    MSM_202209_UP_~ 1.39  0.896      1 mg N~ Ammo~ 1      MSM_20~
## 2 RESULT    MSM_202209_UP_~ 0.681 0.444      1 mg N~ Ammo~ 1      MSM_20~
## 3 RESULT    MSM_202209_UP_~ 0.0299 0.0296      1 mg N~ Ammo~ 1      MSM_20~
## 4 RESULT    MSM_202209_UP_~ 0.194 0.134      1 mg N~ Ammo~ 1      MSM_20~
```

```
## 5 RESULT MSM_202209_UP_~ 0.395 0.262 1 mg N~ Ammo~ 1 MSM_20~
## 6 RESULT MSM_202209_UP_~ 0.375 0.250 1 mg N~ Ammo~ 1 MSM_20~
## # i 1 more variable: Original_Name <chr>
```

```
P04samples <- subset(allldat2, Test == "o-PHOS 0.3")
head(P04samples)
```

```
## # A tibble: 6 x 10
## Run_Info Sample_Name Conc Abs Dilution Units Test Run_Number Pair_ID
## <chr> <chr> <dbl> <dbl> <dbl> <chr> <chr> <chr> <chr>
## 1 RESULT MSM_202209_UP_~ 0.0160 0.00720 1 mg P~ o-PH~ 1 MSM_20~
## 2 RESULT MSM_202209_UP_~ 0.0423 0.0182 1 mg P~ o-PH~ 1 MSM_20~
## 3 RESULT MSM_202209_UP_~ 0.0116 0.00537 1 mg P~ o-PH~ 1 MSM_20~
## 4 RESULT MSM_202209_UP_~ 0.00914 0.00435 1 mg P~ o-PH~ 1 MSM_20~
## 5 RESULT MSM_202209_UP_~ 0.0393 0.0170 1 mg P~ o-PH~ 1 MSM_20~
## 6 RESULT MSM_202209_UP_~ 0.0391 0.0169 1 mg P~ o-PH~ 1 MSM_20~
## # i 1 more variable: Original_Name <chr>
```

```
NOXsamples <- subset(Nalldat2, Test == "Vanadium NOx")
head(NOXsamples)
```

```
## # A tibble: 6 x 10
## Run_Info Sample_Name Conc Abs Dilution Units Test Run_Number Pair_ID
## <chr> <chr> <dbl> <dbl> <dbl> <chr> <chr> <chr> <chr>
## 1 RESULT MSM_202209_UP_~ 5.18e-3 0.189 1 mg N~ Vana~ 1 MSM_20~
## 2 RESULT MSM_202209_UP_~ 4.45e-3 0.189 1 mg N~ Vana~ 1 MSM_20~
## 3 RESULT MSM_202209_UP_~ -6 e-4 0.185 1 mg N~ Vana~ 1 MSM_20~
## 4 RESULT MSM_202209_UP_~ -5.84e-4 0.185 1 mg N~ Vana~ 1 MSM_20~
## 5 RESULT MSM_202209_UP_~ 2.02e-2 0.201 1 mg N~ Vana~ 1 MSM_20~
## 6 RESULT MSM_202209_UP_~ -2.94e-3 0.183 1 mg N~ Vana~ 1 MSM_20~
## # i 1 more variable: Original_Name <chr>
```

```
NH4samples$Conc_uM <- (((as.numeric(NH4samples$Conc))/Con1)/N_mw)*Con2
head(NH4samples)
```

```
## # A tibble: 6 x 11
## Run_Info Sample_Name Conc Abs Dilution Units Test Run_Number Pair_ID
## <chr> <chr> <dbl> <dbl> <dbl> <chr> <chr> <chr> <chr>
## 1 RESULT MSM_202209_UP_~ 1.39 0.896 1 mg N~ Ammo~ 1 MSM_20~
## 2 RESULT MSM_202209_UP_~ 0.681 0.444 1 mg N~ Ammo~ 1 MSM_20~
## 3 RESULT MSM_202209_UP_~ 0.0299 0.0296 1 mg N~ Ammo~ 1 MSM_20~
## 4 RESULT MSM_202209_UP_~ 0.194 0.134 1 mg N~ Ammo~ 1 MSM_20~
## 5 RESULT MSM_202209_UP_~ 0.395 0.262 1 mg N~ Ammo~ 1 MSM_20~
## 6 RESULT MSM_202209_UP_~ 0.375 0.250 1 mg N~ Ammo~ 1 MSM_20~
## # i 2 more variables: Original_Name <chr>, Conc_uM <dbl>
```

```
P04samples$Conc_uM <- (((as.numeric(P04samples$Conc))/Con1)/N_mw)*Con2
head(P04samples)
```

```
## # A tibble: 6 x 11
## Run_Info Sample_Name Conc Abs Dilution Units Test Run_Number Pair_ID
```

```
##   <chr>      <chr>          <dbl>  <dbl>      <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_202209_U~ 0.0160  0.00720      1 mg P~ o-PH~ 1      MSM_20~
## 2 RESULT    MSM_202209_U~ 0.0423  0.0182      1 mg P~ o-PH~ 1      MSM_20~
## 3 RESULT    MSM_202209_U~ 0.0116  0.00537      1 mg P~ o-PH~ 1      MSM_20~
## 4 RESULT    MSM_202209_U~ 0.00914 0.00435      1 mg P~ o-PH~ 1      MSM_20~
## 5 RESULT    MSM_202209_U~ 0.0393  0.0170      1 mg P~ o-PH~ 1      MSM_20~
## 6 RESULT    MSM_202209_U~ 0.0391  0.0169      1 mg P~ o-PH~ 1      MSM_20~
## # i 2 more variables: Original_Name <chr>, Conc_uM <dbl>
```

```
NOXsamples$Conc_uM_raw <- (((as.numeric(NOXsamples$Conc))/Con1)/N_mw)*Con2
head(NOXsamples)
```

```
## # A tibble: 6 x 11
##   Run_Info Sample_Name      Conc  Abs Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>          <dbl> <dbl>      <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_202209_UP~ 5.18e-3 0.189      1 mg N~ Vana~ 1      MSM_20~
## 2 RESULT    MSM_202209_UP~ 4.45e-3 0.189      1 mg N~ Vana~ 1      MSM_20~
## 3 RESULT    MSM_202209_UP~ -6 e-4 0.185      1 mg N~ Vana~ 1      MSM_20~
## 4 RESULT    MSM_202209_UP~ -5.84e-4 0.185      1 mg N~ Vana~ 1      MSM_20~
## 5 RESULT    MSM_202209_UP~ 2.02e-2 0.201      1 mg N~ Vana~ 1      MSM_20~
## 6 RESULT    MSM_202209_UP~ -2.94e-3 0.183      1 mg N~ Vana~ 1      MSM_20~
## # i 2 more variables: Original_Name <chr>, Conc_uM_raw <dbl>
```

```
#add step to make negative values equal to bd (below detection) and replace with zeros
NOXsamples$Conc_uM <- ifelse(NOXsamples$Conc_uM_raw<0, 0, (NOXsamples$Conc_uM_raw) )
head(NOXsamples)
```

```
## # A tibble: 6 x 12
##   Run_Info Sample_Name      Conc  Abs Dilution Units Test Run_Number Pair_ID
##   <chr>      <chr>          <dbl> <dbl>      <dbl> <chr> <chr> <chr>      <chr>
## 1 RESULT    MSM_202209_UP~ 5.18e-3 0.189      1 mg N~ Vana~ 1      MSM_20~
## 2 RESULT    MSM_202209_UP~ 4.45e-3 0.189      1 mg N~ Vana~ 1      MSM_20~
## 3 RESULT    MSM_202209_UP~ -6 e-4 0.185      1 mg N~ Vana~ 1      MSM_20~
## 4 RESULT    MSM_202209_UP~ -5.84e-4 0.185      1 mg N~ Vana~ 1      MSM_20~
## 5 RESULT    MSM_202209_UP~ 2.02e-2 0.201      1 mg N~ Vana~ 1      MSM_20~
## 6 RESULT    MSM_202209_UP~ -2.94e-3 0.183      1 mg N~ Vana~ 1      MSM_20~
## # i 3 more variables: Original_Name <chr>, Conc_uM_raw <dbl>, Conc_uM <dbl>
```

Pull all data back together and add flags

```
#pull out the columns we want from each dataframe
NH4_pull <- NH4samples[ ,c(2,3,11) ]
head(NH4_pull)
```

```
## # A tibble: 6 x 3
##   Sample_Name      Conc Conc_uM
##   <chr>          <dbl>  <dbl>
## 1 MSM_202209_UP_LysA_20cm 1.39    99.3
## 2 MSM_202209_UP_LysA_45cm 0.681   48.6
## 3 MSM_202209_UP_LysB_10cm 0.0299   2.13
```

```
## 4 MSM_202209_UP_LysB_20cm 0.194    13.9
## 5 MSM_202209_UP_LysB_45cm 0.395    28.2
## 6 MSM_202209_UP_LysC_10cm 0.375    26.8
```

```
P04_pull <- P04samples[ ,c(2,3,11) ]
head(P04_pull)
```

```
## # A tibble: 6 x 3
##   Sample_Name      Conc Conc_uM
##   <chr>          <dbl>   <dbl>
## 1 MSM_202209_UP_LysA_20cm 0.0160    1.14
## 2 MSM_202209_UP_LysA_45cm 0.0423    3.02
## 3 MSM_202209_UP_LysB_10cm 0.0116    0.826
## 4 MSM_202209_UP_LysB_20cm 0.00914   0.653
## 5 MSM_202209_UP_LysB_45cm 0.0393    2.81
## 6 MSM_202209_UP_LysC_20cm 0.0391    2.79
```

```
NOX_pull <- NOXsamples[ ,c(2,3,12) ]
head(NOX_pull)
```

```
## # A tibble: 6 x 3
##   Sample_Name      Conc Conc_uM
##   <chr>          <dbl>   <dbl>
## 1 MSM_202209_UP_LysA_20cm 0.00518   0.370
## 2 MSM_202209_UP_LysA_45cm 0.00445   0.318
## 3 MSM_202209_UP_LysB_10cm -0.0006    0
## 4 MSM_202209_UP_LysB_20cm -0.000584 0
## 5 MSM_202209_UP_LysB_45cm 0.0202    1.44
## 6 MSM_202209_UP_LysC_10cm -0.00294  0
```

```
#Bring all this data back together:
all_data <- merge(NH4_pull, P04_pull, by="Sample_Name", all.x=TRUE)
all_data <- merge(all_data, NOX_pull, by="Sample_Name", all.x=TRUE)
head(all_data)
```

```
##           Sample_Name  Conc.x Conc_uM.x  Conc.y Conc_uM.y      Conc
## 1      GCrew_202209_SW_A 0.267645 19.108355 0.065339 4.6648390 0.018448
## 2      GCrew_202209_SW_B 0.174531 12.460537 0.060568 4.3242163 0.004947
## 3      GCrew_202209_SW_C 0.095026  6.784325 0.038633 2.7581800 0.009035
## 4 GCrew_202209_TR_LysA_20cm 0.288862 20.623130 0.007691 0.5490944      NA
## 5 GCrew_202209_TR_LysA_45cm 0.110145  7.863737 0.008317 0.5937873      NA
## 6 GCrew_202209_TR_LysB_10cm 0.454608 32.456467 0.007700 0.5497369 0.006141
##           Conc_uM
## 1 1.3170840
## 2 0.3531881
## 3 0.6450484
## 4      NA
## 5      NA
## 6 0.4384330
```

```
colnames(all_data) <- c("Sample_Name", "NH3_mgL", "NH3_uM", "PO4_mgL", "PO4_uM", "NOx_mgL", "NOx_uM")
head(all_data)
```

```
##           Sample_Name NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL
## 1      GCrew_202209_SW_A 0.267645 19.108355 0.065339 4.6648390 0.018448
## 2      GCrew_202209_SW_B 0.174531 12.460537 0.060568 4.3242163 0.004947
## 3      GCrew_202209_SW_C 0.095026  6.784325 0.038633 2.7581800 0.009035
## 4 GCrew_202209_TR_LysA_20cm 0.288862 20.623130 0.007691 0.5490944      NA
## 5 GCrew_202209_TR_LysA_45cm 0.110145  7.863737 0.008317 0.5937873      NA
## 6 GCrew_202209_TR_LysB_10cm 0.454608 32.456467 0.007700 0.5497369 0.006141
##           NOx_uM
## 1 1.3170840
## 2 0.3531881
## 3 0.6450484
## 4      NA
## 5      NA
## 6 0.4384330
```

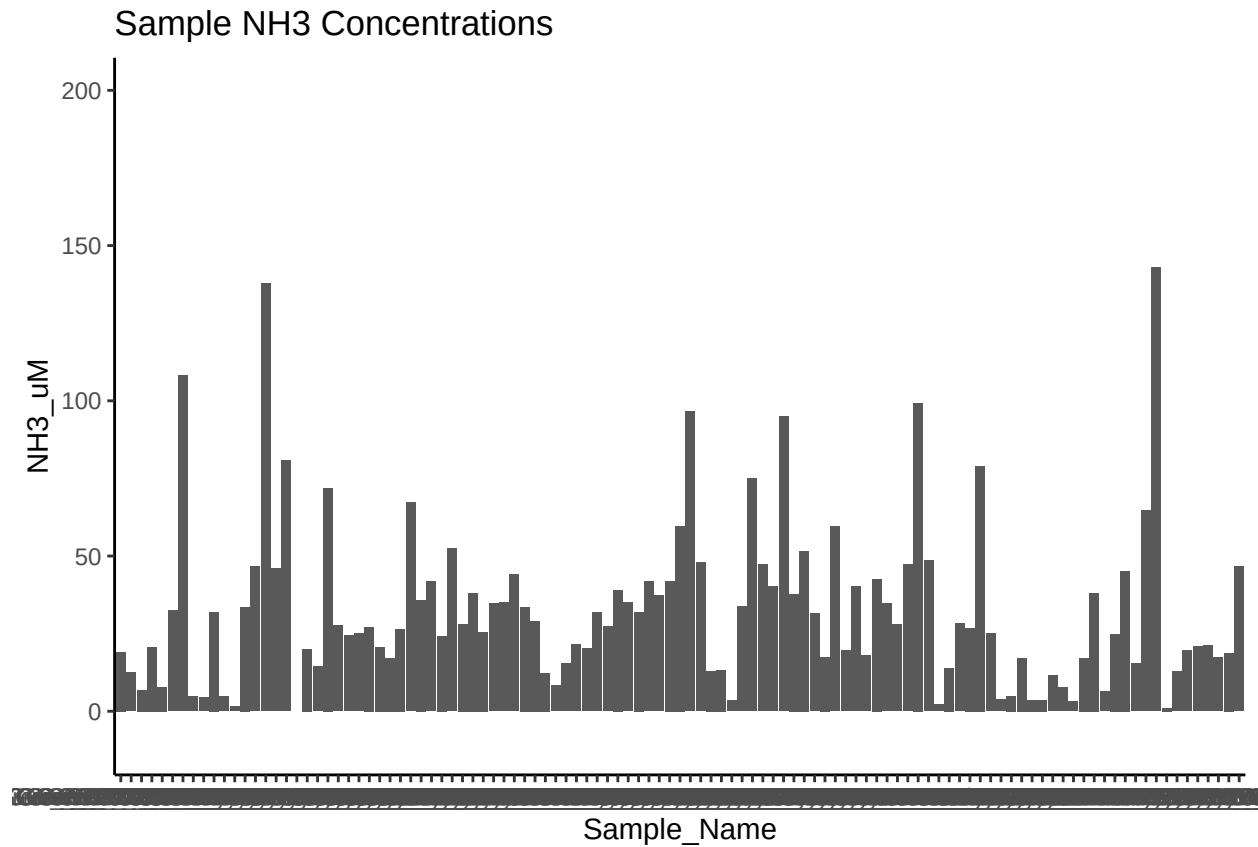
```
#add in an if then statement that tells us if they are within the range of the test - check this after
all_data$NH3_range <- ifelse(all_data$NH3_mgL<0.02, "bd1", ifelse(all_data$NH3_mgL>2, "ad1", "Within_Range"))
all_data$PO4_range <- ifelse(all_data$PO4_mgL<0.003, "bd1", ifelse(all_data$PO4_mgL>3, "ad1", "Within_Range"))
all_data$NOx_range <- ifelse(all_data$NOx_mgL<0.025, "bd1", ifelse(all_data$NOx_mgL>1, "ad1", "Within_Range"))
head(all_data)
```

```
##           Sample_Name NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL
## 1      GCrew_202209_SW_A 0.267645 19.108355 0.065339 4.6648390 0.018448
## 2      GCrew_202209_SW_B 0.174531 12.460537 0.060568 4.3242163 0.004947
## 3      GCrew_202209_SW_C 0.095026  6.784325 0.038633 2.7581800 0.009035
## 4 GCrew_202209_TR_LysA_20cm 0.288862 20.623130 0.007691 0.5490944      NA
## 5 GCrew_202209_TR_LysA_45cm 0.110145  7.863737 0.008317 0.5937873      NA
## 6 GCrew_202209_TR_LysB_10cm 0.454608 32.456467 0.007700 0.5497369 0.006141
##           NOx_uM      NH3_range      PO4_range      NOx_range
## 1 1.3170840 Within_Range Within_Range      bdl
## 2 0.3531881 Within_Range Within_Range      bdl
## 3 0.6450484 Within_Range Within_Range      bdl
## 4      NA Within_Range Within_Range      <NA>
## 5      NA Within_Range Within_Range      <NA>
## 6 0.4384330 Within_Range Within_Range      bdl
```

Take an initial look at concentrations

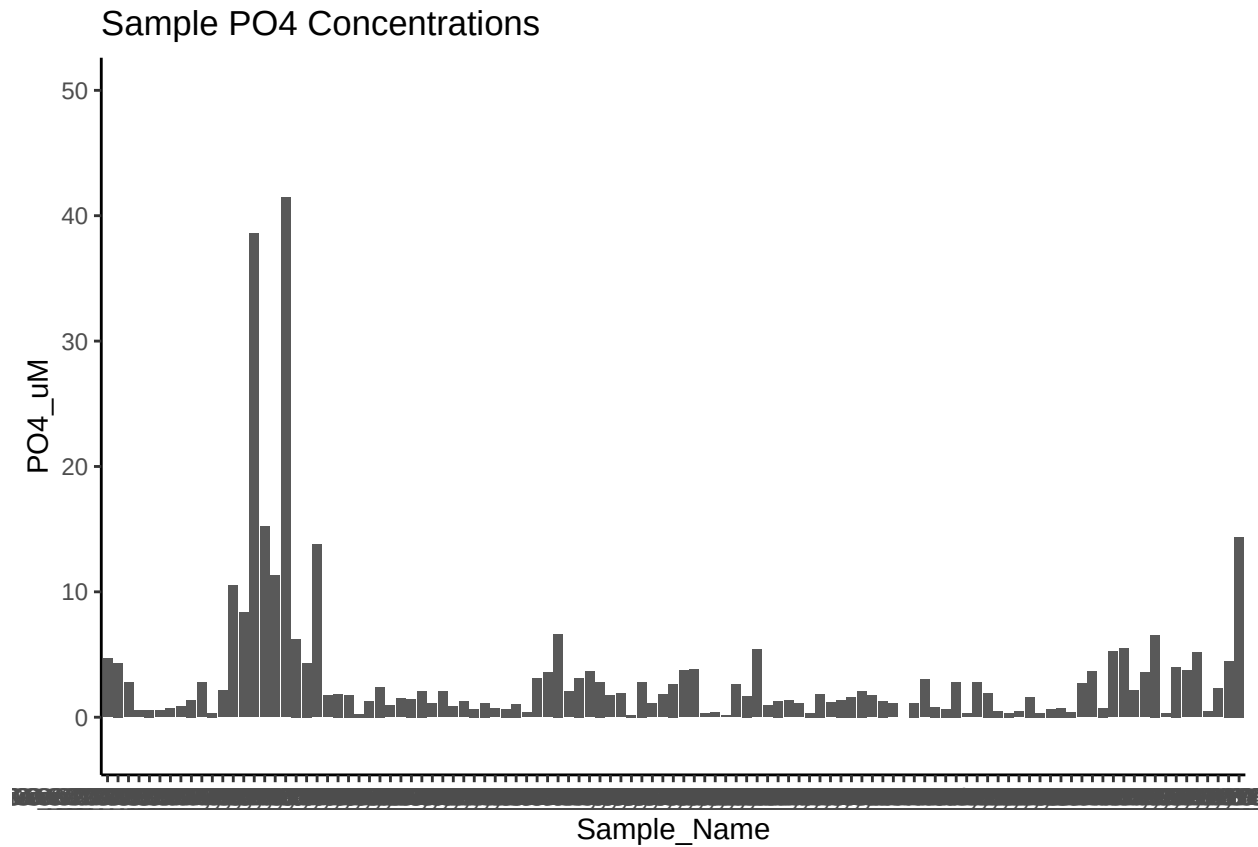
```
#plot data to get a sense of any outliers
NH3look <- ggplot(data=all_data, aes(x=Sample_Name, y=NH3_uM)) +
  geom_bar(stat="identity") +
  theme_classic() + ylim(-10, 200) +
  theme(legend.position="none") +
  ggtitle("Sample NH3 Concentrations")
NH3look
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```



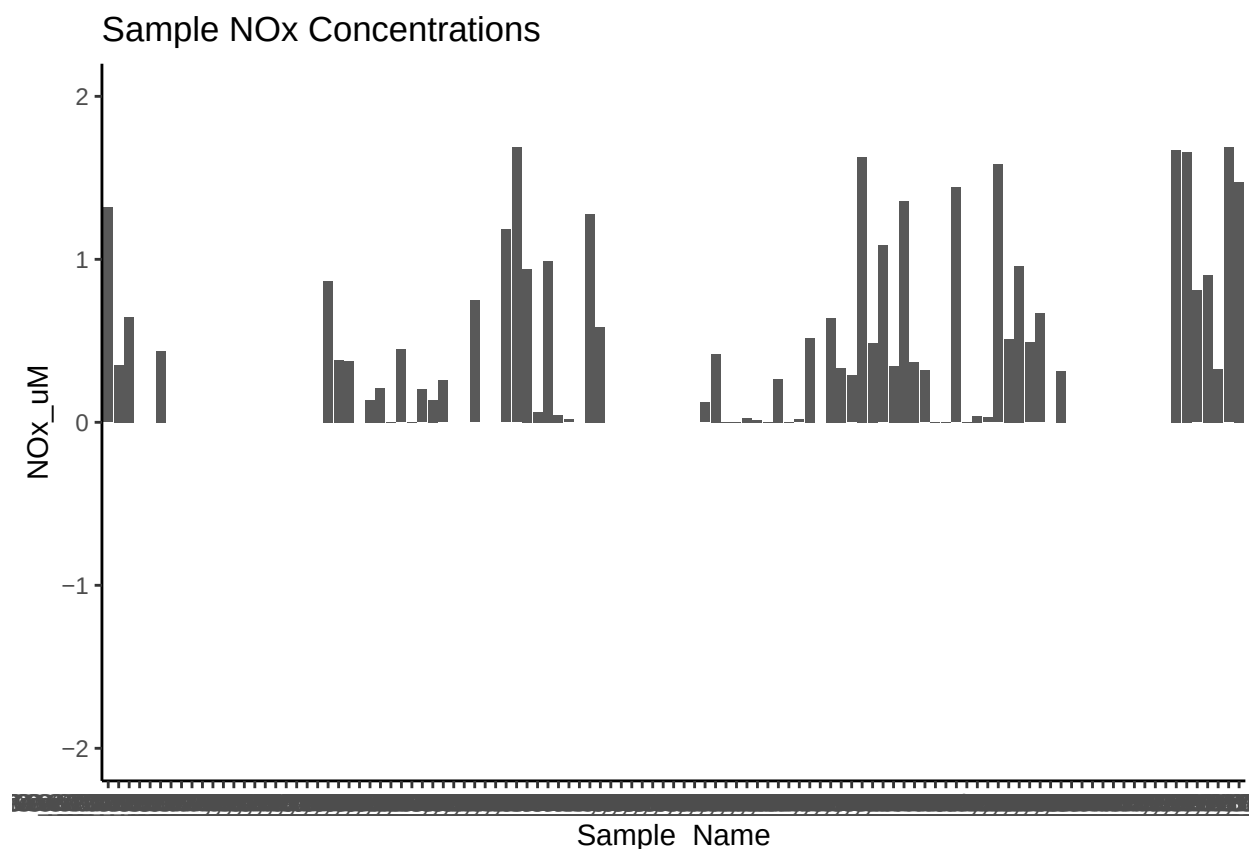
```
P04look <- ggplot(data=all_data, aes(x=Sample_Name, y=P04_uM)) +
  geom_bar(stat="identity") +
  theme_classic() + ylim(-2, 50) +
  theme(legend.position="none") +
  ggtitle("Sample P04 Concentrations")
P04look
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```



```
NOXlook <- ggplot(data=all_data, aes(x=Sample_Name, y=NOx_uM)) +
  geom_bar(stat="identity") +
  theme_classic() + ylim(-2, 2) +
  theme(legend.position="none") +
  ggtitle("Sample NOx Concentrations")
NOXlook
```

```
## Warning: Removed 44 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



Pull out data you need, make IDs

```
head(all_data)
```

```
##           Sample_Name NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL
## 1      GCrew_202209_SW_A 0.267645 19.108355 0.065339 4.6648390 0.018448
## 2      GCrew_202209_SW_B 0.174531 12.460537 0.060568 4.3242163 0.004947
## 3      GCrew_202209_SW_C 0.095026  6.784325 0.038633 2.7581800 0.009035
## 4 GCrew_202209_TR_LysA_20cm 0.288862 20.623130 0.007691 0.5490944      NA
## 5 GCrew_202209_TR_LysA_45cm 0.110145  7.863737 0.008317 0.5937873      NA
## 6 GCrew_202209_TR_LysB_10cm 0.454608 32.456467 0.007700 0.5497369 0.006141
##      NOx_uM  NH3_range  PO4_range NOx_range
## 1 1.3170840 Within_Range Within_Range      bdl
## 2 0.3531881 Within_Range Within_Range      bdl
## 3 0.6450484 Within_Range Within_Range      bdl
## 4      NA Within_Range Within_Range      <NA>
## 5      NA Within_Range Within_Range      <NA>
## 6 0.4384330 Within_Range Within_Range      bdl
```

```
# out <- all_data[,c(1,3,5,7,8,9,10)]
out <- all_data
head(out)
```

```
##           Sample_Name NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL
## 1      GCrew_202209_SW_A 0.267645 19.108355 0.065339 4.6648390 0.018448
## 2      GCrew_202209_SW_B 0.174531 12.460537 0.060568 4.3242163 0.004947
## 3      GCrew_202209_SW_C 0.095026  6.784325 0.038633 2.7581800 0.009035
## 4 GCrew_202209_TR_LysA_20cm 0.288862 20.623130 0.007691 0.5490944      NA
## 5 GCrew_202209_TR_LysA_45cm 0.110145  7.863737 0.008317 0.5937873      NA
## 6 GCrew_202209_TR_LysB_10cm 0.454608 32.456467 0.007700 0.5497369 0.006141
##      NOx_uM  NH3_range  PO4_range NOx_range
## 1 1.3170840 Within_Range Within_Range      bdl
## 2 0.3531881 Within_Range Within_Range      bdl
## 3 0.6450484 Within_Range Within_Range      bdl
## 4      NA Within_Range Within_Range      <NA>
## 5      NA Within_Range Within_Range      <NA>
## 6 0.4384330 Within_Range Within_Range      bdl
```

```
#for steph <- pull out identifiers of the sample names
#pull the sample ID and separate it by the underscores
IDs <- data.frame(do.call('rbind', strsplit(as.character(out$Sample_Name), '_', fixed=TRUE)))
```

```
## Warning in rbind(c("GCrew", "202209", "SW", "A"), c("GCrew", "202209", "SW", :
## number of columns of result is not a multiple of vector length (arg 1)
```

```
colnames(IDs) <- c("Site", "Date", "Zone", "Replicate", "Depth")
IDs$Month <- "September"
head(IDs)
```

```
##      Site  Date Zone Replicate Depth      NA      Month
## 1 GCrew 202209  SW          A GCrew 202209 September
## 2 GCrew 202209  SW          B GCrew 202209 September
## 3 GCrew 202209  SW          C GCrew 202209 September
## 4 GCrew 202209  TR      LysA 20cm GCrew September
## 5 GCrew 202209  TR      LysA 45cm GCrew September
## 6 GCrew 202209  TR      LysB 10cm GCrew September
```

```
IDs$"NA" <- NULL
```

```
#rejoin them to the dataframe
alldat <- cbind(IDs, out)
head(alldat)
```

```
##      Site  Date Zone Replicate Depth      Month      Sample_Name
## 1 GCrew 202209  SW          A GCrew September GCrew_202209_SW_A
## 2 GCrew 202209  SW          B GCrew September GCrew_202209_SW_B
## 3 GCrew 202209  SW          C GCrew September GCrew_202209_SW_C
## 4 GCrew 202209  TR      LysA 20cm September GCrew_202209_TR_LysA_20cm
## 5 GCrew 202209  TR      LysA 45cm September GCrew_202209_TR_LysA_45cm
## 6 GCrew 202209  TR      LysB 10cm September GCrew_202209_TR_LysB_10cm
##      NH3_mgL  NH3_uM  PO4_mgL  PO4_uM  NOx_mgL  NOx_uM  NH3_range
## 1 0.267645 19.108355 0.065339 4.6648390 0.018448 1.3170840 Within_Range
## 2 0.174531 12.460537 0.060568 4.3242163 0.004947 0.3531881 Within_Range
## 3 0.095026  6.784325 0.038633 2.7581800 0.009035 0.6450484 Within_Range
## 4 0.288862 20.623130 0.007691 0.5490944      NA      NA Within_Range
```

```
## 5 0.110145 7.863737 0.008317 0.5937873      NA      NA Within_Range
## 6 0.454608 32.456467 0.007700 0.5497369 0.006141 0.4384330 Within_Range
##      PO4_range NOx_range
## 1 Within_Range      bdl
## 2 Within_Range      bdl
## 3 Within_Range      bdl
## 4 Within_Range      <NA>
## 5 Within_Range      <NA>
## 6 Within_Range      bdl
```

Export final data with flags

```
#Export Data
alldat$Run_notes = Run_notes
write.csv(alldat, file="Processed Data/COMPASS_SynopticCB_Nutrients_202209.csv")
```